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18.02 Multivariable Calculus
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18.02 Lecture 26. – Tue, Nov 13, 2007

Spherical coordinates (ρ, ϕ, θ) .

$\rho = \text{rho}$ = distance to origin. $\phi = \varphi = \text{phi}$ = angle down from z -axis. θ = same as in cylindrical coordinates. Diagram drawn in space, and picture of 2D slice by vertical plane with z, r coordinates.

Formulas to remember: $z = \rho \cos \phi$, $r = \rho \sin \phi$ (so $x = \rho \sin \phi \cos \theta$, $y = \rho \sin \phi \sin \theta$).

$\rho = \sqrt{x^2 + y^2 + z^2} = \sqrt{r^2 + z^2}$. The equation $\rho = a$ defines the sphere of radius a centered at 0.

On the surface of the sphere, ϕ is similar to *latitude*, except it's 0 at the north pole, $\pi/2$ on the equator, π at the south pole. θ is similar to *longitude*.

$\phi = \pi/4$ is a cone (asked using flash cards) ($z = r = \sqrt{x^2 + y^2}$). $\phi = \pi/2$ is the xy -plane.

Volume element: $dV = \rho^2 \sin \phi d\rho d\phi d\theta$.

To understand this formula, first study surface area on sphere of radius a : picture shown of a “rectangle” corresponding to $\Delta\phi$, $\Delta\theta$, with sides = portion of circle of radius a , of length $a\Delta\phi$, and portion of circle of radius $r = a \sin \phi$, of length $r\Delta\theta = a \sin \phi \Delta\theta$. So $\Delta S \approx a^2 \sin \phi \Delta\phi \Delta\theta$, which gives the surface element $dS = a^2 \sin \phi d\phi d\theta$.

The volume element follows: for a small “box”, $\Delta V = \Delta S \Delta\rho$, so $dV = d\rho dS = \rho^2 \sin \phi d\rho d\phi d\theta$.

Example: recall the complicated example at end of Friday's lecture (region sliced by a plane inside unit sphere). After rotating coordinate system, the question becomes: volume of the portion of unit sphere above the plane $z = 1/\sqrt{2}$? (picture drawn). This can be set up in cylindrical (left as exercise) or spherical coordinates.

For fixed ϕ, θ we are slicing our region by rays straight out of the origin; ρ ranges from its value on the plane $z = 1/\sqrt{2}$ to its value on the sphere $\rho = 1$. Spherical coordinate equation of the plane: $z = \rho \cos \phi = 1/\sqrt{2}$, so $\rho = \sec \phi / \sqrt{2}$. The volume is:

$$\int_0^{2\pi} \int_0^{\pi/4} \int_{\frac{1}{\sqrt{2}} \sec \phi}^1 \rho^2 \sin \phi d\rho d\phi d\theta.$$

(Bound for ϕ explained by looking at a slice by vertical plane $\theta = \text{constant}$: the edge of the region is at $z = r = \frac{1}{\sqrt{2}}$).

Evaluation: not done. Final answer: $\frac{2\pi}{3} - \frac{5\pi}{6\sqrt{2}}$.

Application to gravitation.

Gravitational force exerted on mass m at origin by a mass ΔM at (x, y, z) (picture shown) is given by $|\vec{F}| = \frac{G \Delta M m}{\rho^2}$, $\text{dir}(\vec{F}) = \frac{\langle x, y, z \rangle}{\rho}$, i.e. $\vec{F} = \frac{G \Delta M m}{\rho^3} \langle x, y, z \rangle$. (G = gravitational constant).

If instead of a point mass we have a solid with density δ , then we must integrate contributions to gravitational attraction from small pieces $\Delta M = \delta \Delta V$. So

$$\vec{F} = \iiint_R \frac{Gm \langle x, y, z \rangle}{\rho^3} \delta dV, \quad \text{i.e. } z\text{-component is } F_z = Gm \iiint_R \frac{z}{\rho^3} \delta dV, \dots$$

If we can set up to use symmetry, then F_z can be computed nicely using spherical coordinates.

General setup: place the mass m at the origin (so integrand is as above), and place the solid so that the z -axis is an axis of symmetry. Then $\vec{F} = \langle 0, 0, F_z \rangle$ by symmetry, and we have only one

component to compute. Then

$$F_z = Gm \iiint_R \frac{z}{\rho^3} \delta dV = Gm \iiint_R \frac{\rho \cos \phi}{\rho^3} \delta \rho^2 \sin \phi d\rho d\phi d\theta = Gm \iiint_R \delta \cos \phi \sin \phi d\rho d\phi d\theta.$$

Example: Newton's theorem: the gravitational attraction of a spherical planet with uniform density δ is the same as that of the equivalent point mass at its center.

[[Setup: the sphere has radius a and is centered on the positive z -axis, tangent to xy -plane at the origin; the test mass is m at the origin. Then

$$F_z = Gm \iiint_R \frac{z}{\rho^3} \delta dV = Gm \int_0^{2\pi} \int_0^{\pi/2} \int_0^{2a \cos \phi} \delta \cos \phi \sin \phi d\rho d\phi d\theta = \dots = \frac{4}{3} Gm\delta \pi a = \frac{GMm}{a^2}$$

where $M = \text{mass of the planet} = \frac{4}{3}\pi a^3 \delta$. (The bounds for ρ and ϕ need to be explained carefully, by drawing a diagram of a vertical slice with z and r coordinate axes, and the inscribed right triangle with vertices the two poles of the sphere + a point on its surface, the hypotenuse is the diameter $2a$ and we get $\rho = 2a \cos \phi$ for the spherical coordinate equation of the sphere).]]

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